# 1 Physics Period 1, Week 4

1. You walk west 220 m, then north 150 m. What are the magnitude and direction of your displacement vector?

2. An object is moving at 18 $m/s$ at $220^o$ counter clockwise from $x$-axis. Find the $x$- and $y$-components of its velocity.

3. Let $\vec{A} = 15\vec{i} - 40\vec{j}$ and $\vec{B} = 31\vec{j} + 18\vec{k}$. Find $\vec{C}$ such that $\vec{A} + \vec{B} + \vec{C} = \vec{0}$

4. An object is moving in the $x$-direction at 1.3 $m/s$ when it undergoes an acceleration $0.52\vec{j}$ $m/s^2$. Find its velocity vector after 4.4 $s$.

5. You wish to row straight across a 63-m wide river. You can row at a steady 1.3 m/s relative to the water, and the river flows at 0.57 m/s.

   (a) What direction should you head?

   (b) How long it will take you to cross the river?

6. A flock of geese is attempting to migrate due south, but the wind is blowing from the west at 5.1 m/s. If the birds can fly at 7.5 m/s relative to air, what direction should they head?

7. You are sailboarding at at 6.5 $m/s$ when a wind gust hits, lasting 6.3 $s$ accelerating your board at 0.48 $m/s^2$ at $35^o$ to your original direction. Find the magnitude and direction of your displacement during the gust.